Ahmed Hossam Khalil

Backend Software Development Engineer

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EXPERIENCE

noon (E-commerce) Egypt
Software Development Engineer 3 *Feb 2024 – Present*

- Own three high-traffic Go services (Customer Catalog API, Nexus, Wishlist API) that power product listing, product detail, and wishlist flows across all noon markets.
- Shipped 25+ features, including four major launches owned end to end with Product, Frontend, and QA; designed the backend architecture for Gift Registry and led its delivery.
- Cut Redis CPU utilization from 75–80% to 25–30% by migrating catalog caching and simplifying access patterns.
- Halved serialization cost in caching, indexing, and API layers via encoding/json/v2 and struct-level optimizations.
- Fully revamped the Wishlist API after 2.5 years idle: redesigned its structure and code, shipped v3 APIs for the new UI, and cut average latency by 40%; fixed an IDOR vulnerability, a recursion storm, and legacy logic errors.
- Built Bigtable Studio (protobuf-aware production data inspection) and Page Builder, two tools the team uses daily.
- Rewrote the shared Bigtable utility layer and migrated every indexing path onto it with zero flagged issues; introduced pre-commit standards across services.
- Designed the Customer Catalog onboarding program and mentor new hires; code reviewer for every team service (Catalog, Nexus, Wishlist, Indexing).

TECHNICAL SKILLS

Languages & Frameworks: Go, Python, C++, SQL, FastAPI, Django
Backend: Distributed Systems, Microservices, API Development, Caching, Performance Optimization, Pub/Sub, Protobuf
Databases: Google Spanner, Bigtable, BigQuery, MySQL, Redis
Cloud & Tools: Google Cloud Platform (GCP), Docker, Kubernetes, CI/CD, Cloud Monitoring & Logging, Git, Linux

EDUCATION

Suez Canal University Egypt
B.Sc. in Computer Science, Grade: Excellent with Honors *Sep 2019 – Jul 2023*

- ICPC Team Leader (2020–2022): led 20+ competitive programmers and ran weekly training and mock contests.

PROJECTS

Polygon Problem Generator | *Python, Claude API, LLM Agents* github.com/7oSkaaa/polygon-problems-generator

- Built a multi-agent Claude pipeline with separate agents for the statement, solution, validator, checker, and tests; stress-tests them against a brute-force solution and exports packages ready to upload to Polygon.

Etlas | *React Native, Expo, Django* github.com/Etlas-SCU/Etlas-Android

- Built the cross-platform React Native app as its main developer and worked on the backend team's Django REST API; the app guides tourists to Egyptian landmarks in Arabic and English (graduation project, graded Excellent).

ACHIEVEMENTS & LEADERSHIP

- **3× ACPC Finalist** (ICPC regional: 49th, 56th, 76th); **3× Top 50 at ECPC** (ICPC national: 23rd, 26th, 48th).
- **ACPC Regional Problem Setting Lead** (2025–2026): Chief Judge for 11 national CPC editions across 7 countries; tested several global Codeforces rounds.
- **Coach, Buzzrameta**: trained 17 ICPC teams; all 17 qualified for ECPC and 9 for ACPC, one reached the ICPC World Finals, and one won Bronze at ECPC 2026.
- **Owner, CP Circus**: partnered with top competitive programmers in Egypt to give advanced-topic sessions in Arabic for teams preparing for ACPC, and prepared a problem sheet for each session and topic.
- **Instructor, Competitive Programming Sessions**: taught 20+ data structures and algorithms topics to students and ICPC teams, then published the notes and problem sets on GitHub (220+ stars).
- **Leader Coach, Coach Academy** (2023–2024): wrote and taught CP course material and mentored the instructor team.